University of California, San Diego

Online RLHF With Linear Dependence on Reward Scale: A Simplified Bandit Analysis

Honors Thesis in B.S Mathematics–Computer Science

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February 4, 2026

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Chapter 1

Introduction

Chapter 2

Background

In this chapter we review the key concepts and algorithms that underlie our work. We start from a high-level introduction to reinforcement learning and policy-gradient methods, then describe the standard RLHF pipeline and popular preference-based optimization techniques such as Direct Preference Optimization (DPO). We then connect these ideas to preference-based bandits and UCB-style exploration, and finally discuss recent theoretical progress on online RLHF, with a focus on the reward-scale dependence and the SE-POPO algorithm.

2.1 Reinforcement Learning and Policy-Gradient Methods

2.1.1 Markov Decision Processes and Policy Objectives

We begin by establishing the general mathematical framework for sequential decision-making. A Reinforcement Learning (RL) problem is typically formalized as a Markov Decision Process (MDP), defined by the tuple $\mathcal{M} = (\mathcal{S}, \mathcal{A}, \mathcal{P}, r, \rho_0, \gamma)$, where:

- $\mathcal{S}$ is the state space.
- $\mathcal{A}$ is the action space.
- $\mathcal{P} : \mathcal{S} \times \mathcal{A} \to \Delta(\mathcal{S})$ represents the transition dynamics, where $\mathcal{P}(s' \mid s, a)$ denotes the probability of transitioning to state s' given state s and action a. Here, $\Delta(\cdot)$ denotes the probability simplex over a set.
- $r : \mathcal{S} \times \mathcal{A} \to \mathbb{R}$ is the reward function, assumed to be bounded.
- $\rho_0 \in \Delta(\mathcal{S})$ is the initial state distribution.
- $\gamma \in [0, 1)$ is the discount factor determining the effective horizon of the problem.

A (stationary) policy $\pi : \mathcal{S} \to \Delta(\mathcal{A})$ maps states to probability distributions over actions. We denote $\pi(a \mid s)$ as the probability of taking action a in state s. The interaction between the agent and the environment induces an (infinite) trajectory $\tau = (s_0, a_0, s_1, a_1, \dots)$, where $s_0 \sim \rho_0$, $a_t \sim \pi(\cdot \mid s_t)$, and $s_{t+1} \sim \mathcal{P}(\cdot \mid s_t, a_t)$ (Sutton and Barto, 2018).

To make the trajectory distribution precise, for any $T \geq 1$ define the length-T, where T is the horizon length of a RL episode, prefix $\tau_{0:T} := (s_0, a_0, s_1, a_1, \ldots, s_T)$. The probability of observing the prefix $\tau_{0:T}$ under policy π is given by:

$$P_\pi(\tau_{0:T}) \;=\; \rho_0(s_0) \prod_{t=0}^{T-1} \pi(a_t \mid s_t)\, \mathcal{P}(s_{t+1} \mid s_t, a_t). \tag{2.1}$$

We write $\tau \sim P_\pi$ to denote a (random) trajectory generated by this process.

The goal of the agent is to find a policy π that maximizes the expected discounted cumulative reward, often referred to as the expected return:

$$J(\pi) := \mathbb{E}_{\tau \sim P_\pi} \left[\sum_{t=0}^{\infty} \gamma^t r(s_t, a_t) \right]. \tag{2.2}$$

To evaluate a policy, we define the value function $V^\pi : \mathcal{S} \to \mathbb{R}$ and the state-action value function (Q-function) $Q^\pi : \mathcal{S} \times \mathcal{A} \to \mathbb{R}$ as:

$$V^\pi(s) := \mathbb{E}_{\tau \sim P_\pi} \left[\sum_{k=0}^{\infty} \gamma^k r(s_{t+k}, a_{t+k}) \,\middle|\, s_t = s \right], \tag{2.3}$$

$$Q^\pi(s, a) := \mathbb{E}_{\tau \sim P_\pi} \left[\sum_{k=0}^{\infty} \gamma^k r(s_{t+k}, a_{t+k}) \,\middle|\, s_t = s, a_t = a \right]. \tag{2.4}$$

These functions satisfy the celebrated Bellman Expectation Equations:

$$Q^\pi(s, a) = r(s, a) + \gamma \mathbb{E}_{s' \sim \mathcal{P}(\cdot|s,a)} \left[V^\pi(s') \right], \quad \text{where} \quad V^\pi(s) = \mathbb{E}_{a \sim \pi(\cdot|s)} \left[Q^\pi(s, a) \right]. \tag{2.5}$$

Remark on the Bandit Setting. While the MDP framework captures the full complexity of sequential dynamics, the core challenge addressed in this thesis—learning from preference feedback under large reward scales—persists even in the absence of state transitions. The Multi-Armed Bandit (MAB) problem, which is the focus of our analysis in Chapters 3 and 4, can be viewed as a specific instance of an MDP where $|\mathcal{S}| = 1$ and $\gamma = 0$. In this simplified setting, the policy π reduces to a distribution over actions (arms), and the objective simplifies to maximizing the expected immediate reward:

$$J(\pi) = \mathbb{E}_{A \sim \pi} \big[r(A) \big].$$

By isolating the preference learning problem from the dynamics learning problem, we can derive sharper statistical bounds while retaining the essential characteristics of the RLHF alignment process.

2.1.2 Policy Gradient and Actor–Critic Methods

Policy Gradient, Actor–Critic Methods, and KL-Regularization

To optimize the objective $J(\pi)$ in (2.2), we parametrize the policy by $\theta \in \mathbb{R}^d$ and write π_θ. We aim to maximize $J(\pi_\theta)$ via stochastic gradient ascent. A standard tool is the *score function estimator* (log-derivative trick).

Trajectory distribution notation. Recall that a policy π_θ together with the environment $(\rho_0, \mathcal{P})$ induces a trajectory distribution P_{π_θ} over $\tau = (s_0, a_0, s_1, a_1, \dots)$. For any finite prefix $\tau_{0:T} = (s_0, a_0, \dots, s_T)$,

$$P_{\pi_\theta}(\tau_{0:T}) = \rho_0(s_0) \prod_{t=0}^{T-1} \pi_\theta(a_t \mid s_t)\, \mathcal{P}(s_{t+1} \mid s_t, a_t).$$

We write $\tau \sim P_{\pi_\theta}$ to denote a trajectory generated by this process.

Policy gradient theorem. Define the (discounted) return along a trajectory

$$R(\tau) := \sum_{t=0}^{\infty} \gamma^t r(s_t, a_t).$$

Assuming (i) sufficient regularity to interchange differentiation and expectation and (ii) the environment dynamics $(\rho_0, \mathcal{P})$ do not depend on θ, we obtain

$$\begin{aligned}
\nabla_\theta J(\pi_\theta) &= \nabla_\theta \mathbb{E}_{\tau \sim P_{\pi_\theta}}[R(\tau)] = \nabla_\theta \int P_{\pi_\theta}(\tau)\, R(\tau)\, d\tau \\
&= \int \nabla_\theta P_{\pi_\theta}(\tau)\, R(\tau)\, d\tau = \int P_{\pi_\theta}(\tau)\, \nabla_\theta \log P_{\pi_\theta}(\tau)\, R(\tau)\, d\tau \\
&= \mathbb{E}_{\tau \sim P_{\pi_\theta}}[\nabla_\theta \log P_{\pi_\theta}(\tau)\, R(\tau)]\,. \qquad (2.6)
\end{aligned}$$

Moreover, since ρ_0 and $\mathcal{P}$ do not depend on θ,

$$\log P_{\pi_\theta}(\tau_{0:T}) = \log \rho_0(s_0) + \sum_{t=0}^{T-1} \log \pi_\theta(a_t \mid s_t) + \sum_{t=0}^{T-1} \log \mathcal{P}(s_{t+1} \mid s_t, a_t),$$

hence

$$\nabla_\theta \log P_{\pi_\theta}(\tau_{0:T}) = \sum_{t=0}^{T-1} \nabla_\theta \log \pi_\theta(a_t \mid s_t).$$

Using the standard identity that replaces the full return by the *action-value* $Q^{\pi_\theta}(s_t, a_t)$ (the expected future discounted return from (s_t, a_t)), we arrive at the **Policy Gradient Theorem**:

$$\nabla_\theta J(\pi_\theta) = \mathbb{E}_{\tau \sim P_{\pi_\theta}} \left[\sum_{t=0}^{T-1} \nabla_\theta \log \pi_\theta(a_t \mid s_t)\, Q^{\pi_\theta}(s_t, a_t) \right]. \qquad (2.7)$$

In practice, Monte Carlo estimates of this gradient can have high variance. *Actor–Critic* methods mitigate this by learning a critic $V_\phi(s)$ or $Q_\phi(s, a)$ to approximate V^{π_θ} or Q^{π_θ}, typically combined with baselines/advantages to reduce variance while keeping the estimator unbiased (or approximately unbiased).

KL-regularized policy optimization. In RLHF, unconstrained maximization of reward can lead to reward hacking or degradation of the base model. A standard remedy is to penalize deviation from a reference policy π_{ref} via a KL term. The KL-regularized objective is

$$J_{\text{KL}}(\pi) := \mathbb{E}_{\tau \sim P_\pi}\left[\sum_{t=0}^{\infty} \gamma^t \left(r(s_t, a_t) - \beta \log \frac{\pi(a_t \mid s_t)}{\pi_{\text{ref}}(a_t \mid s_t)}\right)\right], \tag{2.8}$$

where $\beta > 0$ is a regularization coefficient. This is a *regularized RL objective* (not a fixed-reward MDP), since the penalty depends on π.

It is often convenient to rewrite (2.8) using discounted state visitation (occupancy) measures. Let

$$d_\pi(s) := (1-\gamma) \sum_{t=0}^{\infty} \gamma^t \, \mathbb{P}_\pi(s_t = s)$$

denote the normalized discounted occupancy distribution. Then (2.8) is equivalent (up to the constant factor $1/(1-\gamma)$) to

$$J_{\text{KL}}(\pi) = \frac{1}{1-\gamma} \, \mathbb{E}_{s \sim d_\pi}\left[\mathbb{E}_{a \sim \pi(\cdot \mid s)}[r(s,a)] - \beta \, \text{KL}(\pi(\cdot \mid s) \,\|\, \pi_{\text{ref}}(\cdot \mid s))\right]. \tag{2.9}$$

Optimal policy form and the role of the soft Q-function. A key result is that the optimal policy for (2.8) has a closed-form Gibbs structure in terms of the *optimal soft action-value function*. Let $Q^*_\beta(s,a)$ denote the optimal action-value under the KL-regularized objective (equivalently, the "soft" Q-function associated with temperature β). For a fixed state s, consider the per-state optimization problem

$$\underset{\pi(\cdot \mid s) \in \Delta(\mathcal{A})}{\text{maximize}} \quad \sum_{a \in \mathcal{A}} \pi(a \mid s) \, Q^*_\beta(s,a) \; - \; \beta \sum_{a \in \mathcal{A}} \pi(a \mid s) \log \frac{\pi(a \mid s)}{\pi_{\text{ref}}(a \mid s)}. \tag{2.10}$$

Forming the Lagrangian and setting first-order conditions yields

$$\pi^*(a \mid s) \propto \pi_{\text{ref}}(a \mid s) \exp\left(\frac{Q^*_\beta(s,a)}{\beta}\right),$$

and therefore

$$\pi^*(a \mid s) = \frac{\pi_{\text{ref}}(a \mid s) \exp\left(\frac{Q^*_\beta(s,a)}{\beta}\right)}{\sum_{a' \in \mathcal{A}} \pi_{\text{ref}}(a' \mid s) \exp\left(\frac{Q^*_\beta(s,a')}{\beta}\right)}. \tag{2.11}$$

Equation (2.11) clarifies that the KL penalty induces a *soft* improvement step relative to π_{ref}.

In the single-step (contextual bandit) setting relevant to RLHF preference learning, the "state" is the prompt x, the "action" is a response y, and the episode terminates after one step. In that case there are no future rewards, so $Q^*_\beta(x,y)$ reduces to the immediate reward $r(x,y)$ (up to a state-dependent constant that cancels in comparisons), and (2.11) becomes the familiar exponential reweighting used in DPO-style derivations.

2.2 Reinforcement Learning from Human Feedback and Direct Preference Optimization

While the general theory of Policy Gradients provides a mechanism for optimizing scalar rewards, modern Large Language Models (LLMs) rely on a specific instantiation of this framework known as Reinforcement Learning from Human Feedback (RLHF). In this section, we review the standard two-stage RLHF pipeline and then derive Direct Preference Optimization (DPO), a method that unifies reward modeling and policy optimization into a single objective.

2.2.1 The Standard RLHF Pipeline

The standard workflow for aligning LLMs(Ouyang et al., 2022), typically consists of three phases: Supervised Fine-Tuning (SFT), Reward Modeling, and Reinforcement Learning.

Let $\mathcal{D} = \{(x, y_w, y_l)\}$ denote a dataset of preferences, where x is a prompt, and y_w, y_l are two generated responses such that human annotators prefer y_w over y_l (denoted $y_w \succ y_l$).

Reward Modeling. The core assumption in RLHF is that human preferences follow the **Bradley–Terry (BT) model**(Bradley and Terry, 1952). We assume there exists a latent reward function $r^*(x, y)$ such that the probability of preferring y_1 over y_2 is given by the logistic function of their reward difference:

$$P(y_1 \succ y_2 \mid x) = \sigma\left(r^*(x, y_1) - r^*(x, y_2)\right), \quad \text{where } \sigma(z) = \frac{1}{1 + e^{-z}}. \tag{2.12}$$

A parameterized reward model $r_\phi(x, y)$ is trained to minimize the negative log-likelihood of the preference data:

$$\mathcal{L}_{\text{RM}}(\phi) = -\mathbb{E}_{(x, y_w, y_l)\sim\mathcal{D}}\left[\log \sigma\left(r_\phi(x, y_w) - r_\phi(x, y_l)\right)\right]. \tag{2.13}$$

Policy Optimization. Once the reward model r_ϕ is fixed, it serves as the reward function for the RL phase. The language model is treated as a policy π_θ, initialized from the SFT model π_{ref}. The optimization objective is the KL-regularized reward maximization discussed in Section 2.1.3:

$$\max_{\pi_\theta} \mathbb{E}_{x\sim\mathcal{D}, y\sim\pi_\theta(\cdot|x)}\left[r_\phi(x, y) - \beta \log \frac{\pi_\theta(y|x)}{\pi_{\text{ref}}(y|x)}\right]. \tag{2.14}$$

Standard approaches typically use Proximal Policy Optimization (PPO)(Schulman et al., 2017) to maximize this objective. However, this two-stage process is often unstable and computationally expensive, as it requires maintaining multiple large models in memory and navigating the complex hyperparameter landscape of Actor-Critic algorithms.

2.2.2 Direct Preference Optimization (DPO)

Direct Preference Optimization (DPO), proposed by (Rafailov et al., 2023), addresses the instability of the two-stage pipeline by leveraging the analytical solution of the KL-regularized objective.

Recall from Equation (2.11) that for a fixed reward function $r(x, y)$, the optimal policy π^* satisfies:

$$\pi^*(y|x) = \frac{1}{Z(x)} \pi_{\text{ref}}(y|x) \exp\left(\frac{r(x, y)}{\beta}\right). \tag{2.15}$$

This equation establishes a bijective mapping between the reward function and the optimal policy. Crucially, we can invert this relationship to express the reward $r(x, y)$ in terms of the optimal policy π^*, the reference policy π_{ref}, and the partition function $Z(x)$:

$$r(x, y) = \beta \log \frac{\pi^*(y|x)}{\pi_{\text{ref}}(y|x)} + \beta \log Z(x). \tag{2.16}$$

This inversion is the key insight of DPO: instead of training a separate reward model r_ϕ, we can treat the policy network π_θ itself as an implicit representation of the reward.

Substituting this expression for $r(x, y)$ into the Bradley–Terry preference model, the probability of preferring y_w over y_l becomes:

$$\begin{aligned} P(y_w \succ y_l \mid x) &= \sigma\left(r(x, y_w) - r(x, y_l)\right) \\ &= \sigma\left(\left(\beta \log \frac{\pi^*(y_w|x)}{\pi_{\text{ref}}(y_w|x)} + \beta \log Z(x)\right) - \left(\beta \log \frac{\pi^*(y_l|x)}{\pi_{\text{ref}}(y_l|x)} + \beta \log Z(x)\right)\right) \\ &= \sigma\left(\beta \log \frac{\pi^*(y_w|x)}{\pi_{\text{ref}}(y_w|x)} - \beta \log \frac{\pi^*(y_l|x)}{\pi_{\text{ref}}(y_l|x)}\right). \end{aligned} \tag{2.17}$$

Notice that the partition function $\beta \log Z(x)$, which is generally intractable to compute, cancels out because it depends only on x, not on the response y.

By replacing the theoretical optimal policy π^* with the parameterized policy π_θ, we obtain the DPO objective. The policy is trained to minimize the negative log-likelihood of the preference data directly, implicitly optimizing the underlying reward:

$$\mathcal{L}_{\text{DPO}}(\pi_\theta; \pi_{\text{ref}}) = -\mathbb{E}_{(x, y_w, y_l) \sim \mathcal{D}}\left[\log \sigma\left(\beta \log \frac{\pi_\theta(y_w|x)}{\pi_{\text{ref}}(y_w|x)} - \beta \log \frac{\pi_\theta(y_l|x)}{\pi_{\text{ref}}(y_l|x)}\right)\right]. \tag{2.18}$$

DPO bypasses the need for an explicit reward model or PPO training loop. It is essentially a classification task where the "logit" is the difference in log-likelihood ratios between the chosen and rejected responses. This simplicity and stability have made DPO the prevailing method for post-training LLMs. However, as we will discuss in subsequent chapters, the theoretical analysis of such preference-based methods often hides the complexity of exploration and reward scaling, motivating our study in the simplified bandit setting.

2.3 Preference-Based Bandits and UCB Exploration

While the DPO framework provides a tractable objective for training policy networks, theoretical analysis of its sample complexity remains challenging due to the complexity of deep neural networks. To rigorously study the statistical limits of learning from preferences, the community often turns to the **Multi-Armed Bandit (MAB)** formalism. Specifically, the problem of learning from pairwise comparisons is known as the **K-armed Dueling Bandit** problem (Yue et al., 2012).

2.3.1 The Dueling Bandit Setting

In the standard Dueling Bandit setting, the learner is given a set of K arms (actions), denoted by $\mathcal{A} = \{1, \ldots, K\}$. At each time step t, the learner selects a pair of arms $(a_i, a_j) \in \mathcal{A} \times \mathcal{A}$ and observes a binary feedback $y_t \in \{0, 1\}$, indicating whether a_i was preferred over a_j.

Unlike the standard MAB setting where the learner observes a scalar reward, here the "reward" is latent. The probability of preference is typically modeled by a utility function $u : \mathcal{A} \to \mathbb{R}$ and a link function σ. A common choice, consistent with the RLHF framework discussed in Section 2.2, is the Bradley–Terry model:

$$P(a_i \succ a_j) = \sigma(u(a_i) - u(a_j)). \tag{2.19}$$

The goal of the learner is to identify the arm with the highest utility u^* (best-arm identification) or to minimize the cumulative regret, defined as the sum of utility gaps between the optimal arm and the selected arms over T rounds.

This setting serves as a rigorous abstraction for Single-Turn RLHF. The "arms" represent the possible responses to a prompt, and the "utility" represents the latent reward $r^*(x, y)$. By studying this simplified setting, we can isolate the statistical efficiency of exploration strategies from the generalization challenges of function approximation.

2.3.2 Upper Confidence Bound (UCB) Exploration

A fundamental challenge in bandit problems is the *exploration-exploitation trade-off*. The learner must balance choosing arms that are empirically known to be good (exploitation) with choosing arms about which there is high uncertainty (exploration).

The most celebrated principle for addressing this is **Optimism in the Face of Uncertainty (OFU)**, typically realized via **Upper Confidence Bound (UCB)** algorithms (Auer et al., 2002). In a utility-based setting, the algorithm maintains an empirical estimate $\hat{\mu}_a(t)$ of the utility (or win rate) for each arm a, along with a confidence interval $C_a(t)$ such that the true parameter lies within $[\hat{\mu}_a(t) - C_a(t), \hat{\mu}_a(t) + C_a(t)]$ with high probability.

The algorithm then selects the arm with the highest upper confidence bound:

$$a_t = \arg\max_{a \in \mathcal{A}} \left(\hat{\mu}_a(t) + C_a(t) \right). \tag{2.20}$$

Standard UCB algorithms often rely on Hoeffding's inequality to construct the confidence width, typically scaling as $C_a(t) \propto \sqrt{\frac{\log T}{N_a(t)}}$, where $N_a(t)$ is the number of times arm a has been sampled.

However, in the context of preference learning and RLHF, the naive application of UCB strategies encounters a critical issue: the non-linearity of the Bradley–Terry model. As we will elaborate in the next section, when the latent rewards (utilities) span a large range, the gradient of the sigmoid function vanishes, making the confidence intervals derived from standard concentration inequalities loose and inefficient. This motivates the need for the specialized self-exploring strategies that are the focus of this thesis.

2.4 Reward-Scale Dependence in Online RLHF

A critical yet often overlooked challenge in Reinforcement Learning from Human Feedback is the scale of the latent reward function. In standard reinforcement learning with scalar rewards, the sample complexity typically scales polynomially with the reward range (e.g., R_{max}). However, in preference-based learning under the Bradley-Terry model, the dependence can become exponential, rendering standard algorithms inefficient for tasks with clear, objective improvements (where reward gaps are large).

Consider the standard approach where an agent learns by comparing its generated response y against a fixed reference response y_{ref}. The probability of the agent winning is given by the logistic function:

$$P(y \succ y_{ref}) = \sigma(r^*(y) - r^*(y_{ref})) = \frac{1}{1 + e^{-(r^*(y) - r^*(y_{ref}))}}. \tag{2.21}$$

The statistical difficulty of estimating the reward difference $r^*(y) - r^*(y_{ref})$ is governed by the derivative of the sigmoid function, which represents the Fisher information of the observation.

When the reward scale R_{max} is small, the sigmoid function operates in its linear region (near 0.5), and preference feedback is highly informative. However, if the current policy finds a response y that is significantly better than the reference (i.e., $r^*(y) \gg r^*(y_{ref})$), the win probability approaches 1. In this regime, the derivative of the sigmoid function decays exponentially:

$$\sigma'(z) = \sigma(z)(1 - \sigma(z)) \approx e^{-|z|} \quad \text{for } |z| \gg 1. \tag{2.22}$$

Consequently, distinguishing between a "good" response y and a "slightly better" response y' becomes statistically difficult if both are compared against a much worse reference y_{ref}. To see this formally, let $\Delta := r^*(y) - r^*(y_{\text{ref}})$ and observe a preference label $Y \in \{0, 1\}$ with

$$\mathbb{P}(Y = 1 \mid \Delta) = \sigma(\Delta).$$

This is a Bernoulli observation with log-likelihood

$$\ell(\Delta; Y) = Y \log \sigma(\Delta) + (1 - Y) \log(1 - \sigma(\Delta)).$$

A direct calculation shows

$$-\mathbb{E}\left[\frac{\partial^2}{\partial \Delta^2} \ell(\Delta; Y)\right] = \sigma(\Delta)\big(1 - \sigma(\Delta)\big) = \sigma'(\Delta),$$

so the Fisher information in Δ equals $\sigma'(\Delta)$. In particular, when $|\Delta| \gg 1$ we have $\sigma'(\Delta) \approx e^{-|\Delta|}$, and therefore estimating Δ to constant accuracy requires on the order of

$$n = \Theta\left(\frac{1}{\sigma'(\Delta)}\right) = \Theta\big(e^{|\Delta|}\big)$$

independent comparisons (up to logarithmic factors). Thus, when reward gaps scale with $R_{\max}$, naive comparisons against a fixed reference can incur sample complexity exponential in $R_{\max}$.

2.5 The Solution: SE-POPO Meta-Algorithm and POPO

To address the exponential dependence on the reward scale described in Section 2.4, recent work has introduced the **Self-Exploring Preference-Incentive Online Preference Optimization (SE-POPO)**(Chen et al., 2025) algorithm. This framework provides the first provably sample-efficient online RLHF algorithm that scales polynomially with the reward range R_{max}, rather than exponentially.

2.5.1 Meta-Algorithm: Dynamic Sampler Updates

The core innovation of SE-POPO is the introduction of a dynamic reference policy, referred to as the *sampler* π_{sam}. In standard online RLHF, the learner compares its generated responses against a fixed reference policy (often the initial SFT model). As the learner improves, the reward gap between the learner and the fixed reference can become arbitrarily large, pushing the win probability $P(y \succ y_{ref})$ into the saturated region of the sigmoid function where gradients vanish.

SE-POPO circumvents this by operating in epochs (or intervals). In each epoch, the algorithm fixes a sampler π_{sam} and runs a subroutine to find a policy that outperforms this sampler. Once the learner sufficiently beats the current sampler, π_{sam} is updated to the learner's current policy. By updating the sampler iteratively $K \approx R_{max}$ times, the algorithm ensures that the reward difference between the learner and the sampler remains bounded (approximately constant). This maintains the optimization process within the "linear" region of the Bradley-Terry model, where preference feedback remains informative.

2.5.2 The POPO Subroutine and Exploration Bonus

Within each epoch, the algorithm executes a subroutine called **POPO (Preference-Incentive Online Preference Optimization)**. Unlike standard approaches that rely on reward-based uncertainty (which is difficult to estimate given only preference feedback), POPO optimizes a regularized objective augmented with a *preference-based exploration bonus*.

The objective takes the general form:

$$\mathcal{L}(\pi) = \mathcal{L}_{DPO}(\pi) + \alpha \cdot G(\pi, \pi_{sam}), \tag{2.23}$$

where $\mathcal{L}_{DPO}$ is the standard likelihood objective, and $G(\pi, \pi_{sam})$ is a bonus term that encourages the policy to explore responses where the preference probability against the sampler is uncertain. In the implementation-friendly version of the algorithm, this bonus simplifies to incentivizing the policy to differ from the reference prior in a controlled manner.

2.5.3 Theoretical Complexity

For the general case of function approximation (e.g., using neural networks), it has been proven that SE-POPO achieves a reward suboptimality gap that scales with $\tilde{O}(\sqrt{R_{max}^5/N})$. While this result successfully establishes a polynomial dependence on R_{max}—a significant breakthrough compared to prior exponential bounds—the high degree of the polynomial reflects the difficulty of exploring in general function spaces. In Chapter 4, we will demonstrate

that for the specific case of the K-armed bandit, this bound can be significantly tightened to a linear dependence.

2.5.4 Remark

Though the 'ladder' algorithm may appear intuitive in retrospect, we emphasize three points: (i) standard RLHF practice uses fixed references, and the exponential penalty was not formally characterized until recently; (ii) theoretical analysis is necessary to determine the optimal update schedule and quantify the sample complexity; (iii) our contribution is proving that in the bandit setting, the dependence on R_max is not merely polynomial but linear, demonstrating that the ladder mechanism is essentially optimal.

Chapter 3

Problem Formulation and Bandit-SE-POPO

In this chapter we formalize the simplified online RLHF problem that will be studied throughout the thesis. We begin by describing the probabilistic model and assumptions, then define the performance metrics of interest, and finally present an optimization viewpoint that interprets our algorithms as regularized preference maximization in policy space.

Unlike the full RLHF model of Section 2.2, we omit the prompt variable and treat each arm a as an abstract 'response.' This removes function approximation issues and allows us to focus on the statistical structure induced by the Bradley–Terry model.

3.1 Probabilistic Model and Assumptions

We consider a K-armed bandit with a finite action set

$$\mathcal{A} = \{1, 2, \ldots, K\}.$$

Each arm $a \in \mathcal{A}$ is associated with a latent reward

$$r^\star(a) \in [0, R_{\max}],$$

for some known upper bound $R_{\max} > 0$. We assume the existence of a unique optimal arm

$$a^\star \in \arg\max_{a \in \mathcal{A}} r^\star(a),$$

and denote the suboptimality gap of arm a by

$$\Delta_a := r^\star(a^\star) - r^\star(a) \geq 0, \qquad \Delta_{\min} := \min_{a \neq a^\star} \Delta_a > 0.$$

The interaction between the learner and the environment is modeled on an underlying probability space $(\Omega, \mathcal{F}, \mathbb{P})$. The learning process unfolds over rounds $t = 1, 2, \ldots, N$. At each round, the learning algorithm selects an ordered pair of arms

$$(A_t^{(1)}, A_t^{(2)}) \in \mathcal{A}^2$$

and observes a stochastic *preference* feedback

$$Y_t \in \{0, 1\},$$

where $Y_t = 1$ indicates that $A_t^{(1)}$ is preferred over $A_t^{(2)}$, and $Y_t = 0$ indicates the opposite. The sequence

$$\big(A_t^{(1)}, A_t^{(2)}, Y_t\big)_{t=1}^T$$

is adapted to the natural filtration

$$\mathcal{F}_t := \sigma\big(A_s^{(1)}, A_s^{(2)}, Y_s : 1 \le s \le t\big), \qquad t = 0, 1, \ldots, N,$$

with the convention $\mathcal{F}_0$ is the trivial σ-algebra.

We assume that preferences are generated according to a Bradley–Terry (BT) model. For any pair of arms $(a, b) \in \mathcal{A}^2$, the conditional probability that a is preferred to b is given by

$$\mathbb{P}\big(Y_t = 1 \mid A_t^{(1)} = a, A_t^{(2)} = b\big) = \sigma\big(r^\star(a) - r^\star(b)\big), \qquad \sigma(z) := \frac{1}{1 + e^{-z}}. \tag{3.1}$$

We also assume conditional independence of the feedback given the chosen arms and the latent rewards.

It is convenient to isolate the martingale noise in the observed preferences. Define

$$\xi_t := Y_t - \mathbb{E}\big[Y_t \mid \mathcal{F}_{t-1}\big], \qquad t = 1, \ldots, N. \tag{3.2}$$

By construction, $(\xi_t)_{t=1}^T$ is a martingale difference sequence (MDS) with respect to $(\mathcal{F}_t)_{t=0}^T$, i.e.,

$$\mathbb{E}\big[\xi_t \mid \mathcal{F}_{t-1}\big] = 0, \quad \forall t,$$

and the increments are uniformly bounded in absolute value by 1. This MDS structure will allow us to apply Bernstein- and Freedman-type concentration inequalities to empirical preference estimates constructed from adaptive data.

Assumption 3.1 (Bounded rewards and unique optimum)**.** The latent reward function $r^\star : \mathcal{A} \to [0, R_{\max}]$ satisfies $0 \le r^\star(a) \le R_{\max}$ for all $a \in \mathcal{A}$, for a known constant $R_{\max} > 0$, and there exists a unique optimal arm $a^\star \in \arg\max_{a \in \mathcal{A}} r^\star(a)$ with minimum gap $\Delta_{\min} > 0$.

3.2 Performance Metrics and Regret

In the online RLHF literature, one often distinguishes between *reward regret*, which measures the loss in latent reward with respect to the optimal policy, and *preference regret*, which measures the deficiency in winning probability against a baseline policy. In our bandit setting, both notions can be defined cleanly.

We first define the reward of a (possibly randomized) policy. A policy is a probability vector

$$\pi \in \Delta_K := \big\{\pi \in \mathbb{R}_+^K : \sum_{a=1}^K \pi(a) = 1\big\},$$

where $\pi(a)$ denotes the probability of choosing arm a as the primary arm $A^{(1)}$. The expected reward of π under $r^\star$ is

$$r^\star(\pi) := \mathbb{E}_{A\sim\pi}\big[r^\star(A)\big] = \sum_{a\in\mathcal{A}} \pi(a) r^\star(a).$$

The reward regret after T rounds is then defined as

$$\mathcal{R}_T^{\text{rew}} := \sum_{t=1}^{T} \big(r^\star(a^\star) - r^\star(\pi_t)\big),$$

where π_t is the (possibly history-dependent) policy used to select $A_t^{(1)}$ at round t.

To define preference regret, we fix a *sampler* distribution $q \in \Delta_K$ which specifies the distribution of the baseline arm $A^{(2)}$. Given a policy π and sampler q, the BT model induces a *win probability*:

$$J(\pi; q) := \mathbb{E}_{A\sim\pi,\, B\sim q}\big[\mathbb{P}\big(A \succ B \mid r^\star\big)\big] = \mathbb{E}_{A\sim\pi,\, B\sim q}\big[\sigma\big(r^\star(A) - r^\star(B)\big)\big]. \tag{3.3}$$

Intuitively, $J(\pi; q)$ is the probability that a sample from π is preferred to a sample from q under the latent reward $r^\star$.

For a fixed sampler q, an optimal policy in this preference sense is

$$\pi_q^\star \in \arg\max_{\pi\in\Delta_K} J(\pi; q). \tag{3.4}$$

The corresponding preference regret after T rounds is

$$\mathcal{R}_T^{\text{pref}}(q) := \sum_{t=1}^{T} \big(J(\pi_q^\star; q) - J(\pi_t; q)\big),$$

where π_t again denotes the policy used at round t. Our analysis will primarily work with preference regret with respect to intermediate samplers q and then relate it back to reward regret via a reduction.

3.3 Algorithm Specification: Bandit-SE-POPO

This section presents a tabular (bandit) instantiation of SE-POPO, adapted to the model in Section 3.1. The motivation follows directly from Section 2.4: if the second response is drawn from a fixed, poor sampler (e.g., π_{ref}), then when the learner becomes much better than the sampler, Bradley–Terry feedback saturates and the effective preference gradient vanishes. SE-POPO resolves this by *self-updating the sampler* so that the learner is always compared against a baseline of comparable quality, keeping comparisons informative.

SE-POPO proceeds over M *intervals* (epochs). In each interval m, we fix a sampler q_m and run the subroutine POPO for T rounds. Note that in the general SE-POPO framework(Section 2.5), the sampler is denoted π_{sam}, but in our bandit setting, we use $q \in \Delta_K$ to denote the sampler distribution, emphasizing its role as a fixed baseline against which the learner's policy π is compared. The output policy of POPO is then used as the sampler for the next interval. Intuitively, each interval climbs a "reward ladder" by a constant amount, and repeating this for $M = O(R_{\max})$ intervals prevents the exponential-in-$R_{\max}$ sample complexity discussed in Section 2.4.

3.3.1 POPO in the Bandit Setting: Objective

In the bandit setting, the policy is a distribution $\pi \in \Delta_K$ over actions $\mathcal{A} = \{1, \dots, K\}$. Let $\pi_{\text{ref}} \in \Delta_K$ be the reference policy, and let $q \in \Delta_K$ be a fixed sampler. At each round t, we draw a pair $(A_t^{(1)}, A_t^{(2)})$ with $A_t^{(1)} \sim \pi_t$ and $A_t^{(2)} \sim q$, observe preference feedback, and label the winner/loser as (A_t^w, A_t^l). Let $\mathcal{D}_t := \{(A_s^w, A_s^l)\}_{s=1}^t$ denote the preference dataset.

Following the implementation-friendly POPO objective, we update the policy by maximizing a DPO-style log-sigmoid likelihood term plus a preference-based exploration bonus. Define the bandit DPO log-likelihood (up to an additive constant)

$$\mathcal{L}_t^{\text{DPO}}(\pi) := \sum_{s=1}^{t} \log \sigma\left(\beta \log \frac{\pi(A_s^w)}{\pi_{\text{ref}}(A_s^w)} - \beta \log \frac{\pi(A_s^l)}{\pi_{\text{ref}}(A_s^l)}\right), \tag{3.5}$$

and the preference-based exploration bonus

$$G(\pi; q) := \mathbb{E}_{a\sim\pi,\, b\sim q}\left[\sigma\left(\beta \log \frac{\pi(a)}{\pi_{\text{ref}}(a)} - \beta \log \frac{\pi(b)}{\pi_{\text{ref}}(b)}\right)\right] = \sum_{a=1}^{K}\sum_{b=1}^{K} \pi(a)\, q(b)\, \sigma\left(\beta \log \frac{\pi(a)}{\pi_{\text{ref}}(a)} - \beta \log \frac{\pi(b)}{\pi_{\text{ref}}(b)}\right). \tag{3.6}$$

The POPO update at round t is then

$$\pi_{t+1} \in \arg\max_{\pi\in\Delta_K} \left\{\mathcal{L}_t^{\text{DPO}}(\pi) + \alpha\, G(\pi; q)\right\}, \tag{3.7}$$

where $\alpha > 0$ controls the strength of preference-based exploration. In the tabular case, (3.7) is a low-dimensional smooth optimization over the simplex and can be solved efficiently (e.g., projected gradient / mirror descent).

3.3.2 Meta-Algorithm: Bandit-SE-POPO

Algorithm 1 Bandit-SE-POPO (tabular instantiation)

Input: Action set $\mathcal{A} = \{1, \dots, K\}$, reference policy π_{ref}, reward scale $R_{\max}$.
Parameters: intervals M (typically $M = O(R_{\max})$), inner horizon T, exploration weight α.
Initialize: sampler $q_1 \leftarrow \pi_{\text{ref}}$.
for $m = 1, \dots, M$ **do**
  $\hat{\pi}_m \leftarrow$ Bandit-POPO$(\pi_{\text{ref}}, q_m, T, \alpha)$
  $q_{m+1} \leftarrow \hat{\pi}_m$ {self-update the sampler}
end for
Return: $\hat{\pi}_M$.

3.3.3 Subroutine: Bandit-POPO

Algorithm 2 Bandit-POPO (Preference-Incentive Online Preference Optimization, tabular)

Input: reference π_{ref}, sampler q, iterations T, exploration weight α.
Initialize: $\pi_1 \leftarrow \pi_{\text{ref}}$, dataset $\mathcal{D}_0 \leftarrow \emptyset$.
for $t = 1, \ldots, T$ **do**
  Sample: $A_t^{(1)} \sim \pi_t, \quad A_t^{(2)} \sim q$.
  Feedback: observe preference label $Y_t \in \{0, 1\}$ and set $(A_t^w, A_t^l) = (A_t^{(1)}, A_t^{(2)})$ if $Y_t = 1$, else $(A_t^{(2)}, A_t^{(1)})$.
  Update dataset: $\mathcal{D}_t \leftarrow \mathcal{D}_{t-1} \cup \{(A_t^w, A_t^l)\}$.
  Policy update: compute π_{t+1} by solving (3.7) using $\mathcal{D}_t$.
end for
Return: averaged policy $\bar{\pi} := \text{Uniform}(\pi_1, \ldots, \pi_{T+1})$.

With the algorithm specified, Chapter 4 shows how the interval-based sampler update prevents preference saturation and yields sample complexity polynomial (and in our bandit setting, linear) in $R_{\max}$.

Chapter 4

Theoretical Analysis: Linear Dependence on the Reward Scale

This chapter provides a rigorous sample-complexity analysis of a tabular (bandit) instantiation of the SE-POPO idea. The central phenomenon motivating our analysis is preference saturation under the Bradley–Terry model: when comparisons are made against a fixed, significantly worse reference, the win probability approaches 0 or 1 and learning can become statistically inefficient. We show that *self-updating the sampler* avoids this pathology and yields a total sample complexity that scales *linearly* in the reward range $R_{\max}$.

A deterministic-sampler specialization. To make the theory precise and avoid distributional subtleties, we analyze a deterministic-sampler specialization of Bandit-SE-POPO. In epoch m, the sampler is a single action (arm) $b_m \in \{1, \ldots, K\}$, and each comparison is between a learner-chosen arm A_t and the fixed baseline arm b_m. Equivalently, this corresponds to choosing $q_m = \delta_{b_m}$ in the dueling-bandit model from Section 3.1. This specialization captures the essential SE-POPO mechanism (moving the reference point) while permitting a clean, fully rigorous ladder argument.

4.1 Main Result

Let $r^*(a) \in [0, R_{\max}]$ denote the latent reward of arm $a \in \{1, \ldots, K\}$ (Assumption 3.1), and let

$$\mathbb{P}(a \succ b) = \sigma\big(r^*(a) - r^*(b)\big)$$

denote the Bradley–Terry win probability.

We analyze the following two-phase procedure: (i) a *coarse ladder phase* that repeatedly finds an arm that beats the current baseline by a constant margin and updates the baseline, and (ii) a *final refinement phase* that identifies an ϵ-optimal arm once the baseline is within a constant distance of optimality.

Theorem 4.1 (Linear dependence on $R_{\max}$)**.** *Consider the K-armed dueling bandit problem under the Bradley–Terry model with rewards in $[0, R_{\max}]$. Fix $\delta \in (0, 1)$ and $\epsilon \in (0, 1)$.*

There exists a deterministic-sampler instantiation of Bandit-SE-POPO that returns an arm $\hat{a}$ satisfying

$$r^*(\hat{a}) \;\geq\; \max_a r^*(a) \;-\; \epsilon$$

with probability at least $1-\delta$, using a total number of preference queries

$$N \;=\; \tilde{O}\left(KR_{\max} \;+\; \frac{K}{\epsilon^2}\right),$$

where $\tilde{O}(\cdot)$ hides logarithmic factors in K, $1/\delta$, and (for the refinement phase) $1/\epsilon$.

Significance. Theorem 4.1 establishes a total sample complexity that is *linear* in $R_{\max}$. This stands in contrast to passive comparison schemes against a fixed reference, which can suffer exponential-in-$R_{\max}$ inefficiency due to preference saturation (Section 2.4). The proof formalizes the intuition that self-updating the sampler converts a globally ill-conditioned problem into a sequence of locally well-conditioned ones.

4.2 Proof Overview

The proof has three ingredients:

1. A *constant preference margin* lemma: if an arm is better than the baseline by at least 1 in reward, then it beats the baseline with probability at least $\sigma(1)$, which is bounded away from 1/2 by a constant.

2. A *fixed-baseline identification* lemma: with $\tilde{O}(K)$ comparisons, we can identify an arm whose win probability against the baseline is within a constant of optimal.

3. A *reward-ladder* lemma: a constant win-probability advantage implies a constant *reward* improvement, so the baseline reward increases by a constant each epoch until it is within 1 of optimality.

After reaching the local regime (baseline within 1 of optimal), we perform a final identification step that achieves ϵ-accuracy with $\tilde{O}(K/\epsilon^2)$ comparisons.

4.3 Step 1: Fixed Baseline Identification in the Preference Domain

Fix a baseline arm $b \in \{1, \ldots, K\}$. For each arm a, define its win probability against the baseline

$$p_b(a) \;:=\; \mathbb{P}(a \succ b) \;=\; \sigma\big(r^*(a) - r^*(b)\big).$$

Since $\sigma(\cdot)$ is strictly increasing, maximizing $p_b(a)$ is equivalent to maximizing $r^*(a)$. Thus, identifying an arm with near-maximal win probability against b is a valid route to identifying a near-optimal reward arm.

We analyze a simple, fully explicit subroutine: sample each arm a exactly n times against baseline b, compute the empirical win rate $\hat{p}_b(a)$, and return an arm with maximal $\hat{p}_b(a)$. This yields clean finite-sample bounds by Hoeffding's inequality. We introduce a tolerance parameter $\eta_{pref} \in (0, 1/2)$ that controls the desired accuracy in preference probability. Lemma 4.2 shows that achieving η_{pref}-accuracy requires $O(K/\eta_{pref})$ samples

Lemma 4.2 (Preference best-arm identification against a fixed baseline)**.** *Fix a baseline arm b and let $\eta_{\text{pref}} \in (0, 1/2)$. Let $\hat{a}$ be chosen by sampling each arm a exactly*

$$n \;\geq\; \frac{2}{\eta_{\text{pref}}^2} \log\left(\frac{2K}{\delta'}\right)$$

times against b and returning $\hat{a} \in \arg\max_a \hat{p}_b(a)$. Then with probability at least $1 - \delta'$,

$$p_b(\hat{a}) \;\geq\; \max_a p_b(a) \;-\; \eta_{\text{pref}}.$$

The total number of comparisons used is $Kn = O\left(\frac{K}{\eta_{\text{pref}}^2} \log \frac{K}{\delta'}\right)$.

Proof. For each arm a, $\hat{p}_b(a)$ is the empirical mean of n i.i.d. Bernoulli trials with mean $p_b(a)$. By Hoeffding's inequality,

$$\mathbb{P}\big(|\hat{p}_b(a) - p_b(a)| > \tfrac{1}{2}\eta_{\text{pref}}\big) \;\leq\; 2\exp\left(-\frac{n\eta_{\text{pref}}^2}{2}\right).$$

By a union bound over $a \in \{1, \ldots, K\}$, with probability at least $1-\delta'$ we have $|\hat{p}_b(a) - p_b(a)| \leq \frac{1}{2}\eta_{\text{pref}}$ for all a. On this event,

$$p_b(\hat{a}) \;\geq\; \hat{p}_b(\hat{a}) - \tfrac{1}{2}\eta_{\text{pref}} \;\geq\; \hat{p}_b(a_b^\star) - \tfrac{1}{2}\eta_{\text{pref}} \;\geq\; p_b(a_b^\star) - \eta_{\text{pref}},$$

where $a_b^\star \in \arg\max_a p_b(a)$. □

4.4 Step 2: A Constant Preference Margin Implies a Constant Reward Gain

Define the constant

$$c_0 \;:=\; \sigma(1) \;-\; \frac{1}{2} \;>\; 0.$$

The next lemma states that if the optimal arm is at least 1 better than the baseline in reward, then there is a *constant* preference margin (bounded away from 0) between the best arm and the baseline.

Lemma 4.3 (Unit-margin preference gap against a baseline)**.** *Fix a baseline arm b and let $a^\star \in \arg\max_a r^*(a)$ be an optimal arm. If $r^*(a^\star) \geq r^*(b) + 1$, then*

$$\max_a p_b(a) \;\geq\; p_b(a^\star) \;\geq\; \sigma(1) \;=\; \frac{1}{2} + c_0, \qquad \textit{and} \qquad p_b(b) \;=\; \frac{1}{2}.$$

In particular, $\max_a p_b(a) - p_b(b) \geq c_0$.

Proof. If $r^*(a^\star) \geq r^*(b) + 1$, then

$$p_b(a^\star) = \sigma\big(r^*(a^\star) - r^*(b)\big) \;\geq\; \sigma(1) \;=\; \frac{1}{2} + c_0$$

since σ is increasing. Also $p_b(b) = \sigma(0) = 1/2$. □

Next, we convert a *preference* advantage into a *reward* improvement. This step is where deterministic baselines are crucial: the relationship is exact because the win probability is a logistic function of the reward difference.

Lemma 4.4 (Preference advantage implies reward gain)**.** *Fix a baseline arm b. If an arm a satisfies $p_b(a) \geq \frac{1}{2} + \eta$ for some $\eta \in (0, 1/2)$, then*

$$r^*(a) - r^*(b) \;\geq\; \log\left(\frac{\frac{1}{2} + \eta}{\frac{1}{2} - \eta}\right) \;:=\; g(\eta) \;>\; 0.$$

Proof. $p_b(a) = \sigma(r^*(a) - r^*(b)) \geq \frac{1}{2} + \eta$ implies

$$\frac{1}{1 + \exp(-(r^*(a) - r^*(b)))} \;\geq\; \frac{1}{2} + \eta.$$

Rearranging yields $\exp(-(r^*(a) - r^*(b))) \leq \frac{\frac{1}{2} - \eta}{\frac{1}{2} + \eta}$, equivalently $r^*(a) - r^*(b) \geq \log\left(\frac{\frac{1}{2} + \eta}{\frac{1}{2} - \eta}\right)$. □

For later use, define the constant reward step size

$$g_0 \;:=\; g\Big(\frac{c_0}{2}\Big) \;=\; \log\left(\frac{\frac{1}{2} + \frac{c_0}{2}}{\frac{1}{2} - \frac{c_0}{2}}\right) \;>\; 0.$$

All constants c_0 and g_0 are numerical and independent of $R_{\max}$.

4.5 Step 3: The Reward Ladder and Total Sample Complexity

We now define the ladder phase precisely. Let b_1 be the initial baseline (e.g., any arm in the support of π_{ref}). For $m = 1, 2, \ldots, M$, run the fixed-baseline identification subroutine of Lemma 4.2 with baseline b_m and tolerance $\eta_{\text{pref}} = c_0/2$, and let $\hat{a}_m$ be the returned arm. Update the baseline by $b_{m+1} \leftarrow \hat{a}_m$.

Lemma 4.5 (Reward ladder progress)**.** *Fix an epoch m and baseline b_m. Suppose $r^*(a^\star) \geq r^*(b_m) + 1$ where $a^\star$ is an optimal arm. If the subroutine returns $\hat{a}_m$ satisfying*

$$p_{b_m}(\hat{a}_m) \;\geq\; \max_a p_{b_m}(a) \;-\; \frac{c_0}{2},$$

then the baseline reward increases by a constant:

$$r^*(b_{m+1}) - r^*(b_m) \;=\; r^*(\hat{a}_m) - r^*(b_m) \;\geq\; g_0.$$

Proof. By Lemma 4.3, $\max_a p_{b_m}(a) \geq \frac{1}{2} + c_0$. Thus

$$p_{b_m}(\hat{a}_m) \;\geq\; \left(\frac{1}{2} + c_0\right) - \frac{c_0}{2} \;=\; \frac{1}{2} + \frac{c_0}{2}.$$

Applying Lemma 4.4 with $\eta = c_0/2$ yields $r^*(\hat{a}_m) - r^*(b_m) \geq g_0$. □

The ladder phase continues until the baseline is within 1 of optimality. Since $r^*(\cdot) \in [0, R_{\max}]$, the baseline can increase by at most $R_{\max}$ in total. Therefore, after at most

$$M \;:=\; \left\lceil \frac{R_{\max}}{g_0} \right\rceil$$

epochs, either the process has already reached an optimal arm, or it has reached a baseline b_{M+1} such that

$$r^*(a^\star) - r^*(b_{M+1}) \;<\; 1.$$

4.5.1 Final refinement to ϵ-optimality

Once the baseline is within 1 of optimality, the best arm's reward is within 1 of the baseline. In this local regime, the logistic link is well-conditioned on the interval $[0, 1]$, since $\sigma'(z)$ is bounded below by $\sigma'(1) > 0$ for $z \in [0, 1]$. We exploit this by identifying an arm whose win probability against the baseline is sufficiently close to the maximum.

Let $b = b_{M+1}$ be the final baseline after the ladder phase, and let $a^\star$ be an optimal arm. Then $\Delta^\star := r^*(a^\star) - r^*(b) \in [0, 1)$ and hence $p_b(a^\star) = \sigma(\Delta^\star) \in [1/2, \sigma(1))$.

Lemma 4.6 (ϵ-optimality from local preference accuracy)**.** *Assume $r^*(a^\star) - r^*(b) \in [0, 1]$ for some baseline arm b. Let $\hat{a}$ satisfy*

$$p_b(\hat{a}) \;\geq\; \max_a p_b(a) \;-\; \sigma'(1)\,\epsilon.$$

Then $r^*(\hat{a}) \geq r^*(a^\star) - \epsilon$.

Proof. Let $\Delta(a) := r^*(a) - r^*(b)$, so that $p_b(a) = \sigma(\Delta(a))$. By assumption, $\Delta(a^\star) \in [0, 1]$. If $\hat{a}$ is optimal (i.e., $r^*(\hat{a}) = r^*(a^\star)$), the result is immediate, so assume $\hat{a}$ is suboptimal, meaning $\Delta(\hat{a}) < \Delta(a^\star)$.

We first establish a lower bound on $\Delta(\hat{a})$. Since $\max_a p_b(a) \geq p_b(a^\star) \geq p_b(b) = \frac{1}{2}$, the assumed preference accuracy gives

$$p_b(\hat{a}) \;\geq\; \max_a p_b(a) - \sigma'(1)\epsilon \;\geq\; \tfrac{1}{2} - \sigma'(1)\epsilon.$$

For $\epsilon \in (0, 1)$, we have $\sigma'(1)\epsilon < \sigma'(1) \approx 0.196$, so $p_b(\hat{a}) > 0.3 > \sigma(-1) \approx 0.269$. Since σ is strictly increasing, this implies $\Delta(\hat{a}) > -1$.

By the mean value theorem applied to σ on the interval $[\Delta(\hat{a}), \Delta(a^\star)]$, there exists $\xi \in (\Delta(\hat{a}), \Delta(a^\star))$ such that

$$\sigma(\Delta(a^\star)) - \sigma(\Delta(\hat{a})) = \sigma'(\xi)\,(\Delta(a^\star) - \Delta(\hat{a})).$$

Since $\Delta(\hat{a}) > -1$ and $\Delta(a^\star) \leq 1$, we have $\xi \in (-1, 1)$. The function $\sigma'(z) = \sigma(z)(1-\sigma(z))$ is symmetric about $z = 0$ and strictly decreasing in $|z|$, so for $\xi \in (-1, 1)$ we have $\sigma'(\xi) > \sigma'(1)$. Therefore,

$$\Delta(a^\star) - \Delta(\hat{a}) \;<\; \frac{\sigma(\Delta(a^\star)) - \sigma(\Delta(\hat{a}))}{\sigma'(1)} \;=\; \frac{p_b(a^\star) - p_b(\hat{a})}{\sigma'(1)} \;\leq\; \frac{\sigma'(1)\epsilon}{\sigma'(1)} \;=\; \epsilon,$$

where the last inequality uses the assumed preference accuracy and $p_b(a^\star) = \max_a p_b(a)$. Since $\Delta(a) = r^*(a) - r^*(b)$, this implies $r^*(a^\star) - r^*(\hat{a}) < \epsilon$, completing the proof. □

To achieve the preference accuracy in Lemma 4.6, we invoke Lemma 4.2 with $\eta_{\text{pref}} = \sigma'(1)\epsilon$.

4.6 Proof of Theorem 4.1

Proof. We split the failure probability as $\delta = \delta_{\text{ladder}} + \delta_{\text{refine}}$ with $\delta_{\text{ladder}} = \delta_{\text{refine}} = \delta/2$.

Ladder phase. Run $M = \lceil R_{\max}/g_0 \rceil$ epochs. In each epoch, apply Lemma 4.2 with baseline b_m, tolerance $\eta_{\text{pref}} = c_0/2$, and failure probability $\delta' = \delta_{\text{ladder}}/M$. By a union bound, with probability at least $1 - \delta_{\text{ladder}}$ all epochs satisfy the required preference guarantee. Condition on this event. Whenever $r^*(a^\star) \geq r^*(b_m) + 1$, Lemma 4.5 implies $r^*(b_{m+1}) \geq r^*(b_m) + g_0$. Since rewards are bounded by $R_{\max}$, after at most M epochs we must reach a baseline $b = b_{M+1}$ such that $r^*(a^\star) - r^*(b) < 1$.

By Lemma 4.2, each epoch uses $O\left(K \log \frac{KM}{\delta_{\text{ladder}}}\right)$ samples (since $\eta_{\text{pref}} = c_0/2$ is a constant), so the ladder phase uses

$$N_{\text{ladder}} = \tilde{O}(KM) = \tilde{O}(KR_{\max}).$$

Refinement phase. With the baseline b satisfying $r^*(a^\star) - r^*(b) < 1$, run Lemma 4.2 one more time with tolerance $\eta_{\text{pref}} = \sigma'(1)\epsilon$ and failure probability δ_{refine}. This produces an arm $\hat{a}$ such that $p_b(\hat{a}) \geq \max_a p_b(a) - \sigma'(1)\epsilon$ with probability at least $1 - \delta_{\text{refine}}$. On this event, Lemma 4.6 yields $r^*(\hat{a}) \geq r^*(a^\star) - \epsilon$.

The refinement phase uses

$$N_{\text{refine}} = O\left(\frac{K}{\sigma'(1)^2\epsilon^2} \log \frac{K}{\delta_{\text{refine}}}\right) = \tilde{O}\left(\frac{K}{\epsilon^2}\right).$$

Total complexity. By a union bound over the ladder and refinement events, the overall success probability is at least $1 - \delta$. Summing sample counts gives

$$N = N_{\text{ladder}} + N_{\text{refine}} = \tilde{O}\left(KR_{\max} + \frac{K}{\epsilon^2}\right),$$

which completes the proof. □

4.7 Discussion

The analysis makes precise why self-updating the sampler eliminates the exponential dependence on $R_{\max}$ that can arise under Bradley–Terry feedback. Rather than attempting to resolve tiny gradients in a saturated regime, the ladder phase only requires distinguishing a *constant* preference advantage (bounded away from 0), which costs $\tilde{O}(K)$ samples per epoch and yields a constant reward improvement. Repeating for $O(R_{\max})$ epochs yields linear dependence on $R_{\max}$. Once the baseline is within a constant distance of optimality, a standard best-arm identification argument in the preference domain yields ϵ-accuracy with $\tilde{O}(K/\epsilon^2)$ additional comparisons.

Chapter 5

Experiments and Empirical Validations

This chapter presents empirical validations of the theoretical results established in Chapter 4. We conduct a series of experiments on synthetic K-armed dueling bandits under the Bradley–Terry model to verify the key claims of Theorem 4.1: (i) sample complexity scales *linearly* with the reward scale $R_{\max}$, (ii) sample complexity scales linearly with the number of arms K, and (iii) the refinement phase exhibits the expected $O(1/\epsilon^2)$ dependence on the target accuracy.

5.1 Experimental Setup

Reward Structure. For all experiments, we use a linearly spaced reward structure where the latent reward of arm $a \in \{1, \ldots, K\}$ is given by

$$r^*(a) = R_{\max} \cdot \frac{a-1}{K-1},$$

so that rewards span the interval $[0, R_{\max}]$ uniformly. This structure ensures a unique optimal arm ($a^* = K$) and provides a controlled setting to study the effect of the reward scale.

Preference Feedback. At each query, the learner selects a pair of arms (a, b) and observes binary feedback $Y \in \{0, 1\}$ according to the Bradley–Terry model:

$$\mathbb{P}(Y = 1 \mid a, b) = \sigma(r^*(a) - r^*(b)),$$

where $\sigma(z) = 1/(1 + e^{-z})$ is the logistic function.

Algorithms. We compare three algorithms:

(i) **Bandit-SE-POPO (Ours):** The deterministic-sampler specialization of SE-POPO analyzed in Chapter 4, which self-updates the baseline arm across epochs.

(ii) **Fixed-Baseline UCB:** A UCB-style algorithm that always compares against a fixed baseline arm (arm 1, with reward 0). This serves as a representative of methods that do not update the reference.

(iii) **Uniform Exploration:** A non-adaptive baseline that samples all $\binom{K}{2}$ pairs uniformly and estimates arm quality via Borda count.

Implementation Details. All experiments use failure probability $\delta = 0.1$ unless otherwise specified. Results are averaged over multiple independent trials, with error bars representing one standard deviation. The theoretical constants from Chapter 4 are:

$$\sigma(1) \approx 0.731, \qquad c_0 = \sigma(1) - \tfrac{1}{2} \approx 0.231,$$
$$\sigma'(1) \approx 0.197, \qquad g_0 \approx 0.47.$$

5.2 Experiment 1: Linear Dependence on Reward Scale

The first experiment validates the central claim of this thesis: Bandit-SE-POPO achieves sample complexity that scales *linearly* with the reward scale $R_{\max}$, in contrast to fixed-baseline methods which can suffer from preference saturation.

Setup. We fix $K = 10$ arms and $\epsilon = 0.5$, and vary $R_{\max} \in \{1, 2, 4, 8, 16, 32\}$. For each configuration, we run 50 independent trials and record the total number of preference queries required to identify an ϵ-optimal arm.

Results. Table 5.1 and Figure 5.1 summarize the results.

Table 5.1: Sample complexity comparison across reward scales $R_{\max}$ with $K = 10$ and $\epsilon = 0.5$. Fixed-Baseline UCB hits the query cap (100,000) and fails completely for large $R_{\max}$.

$R_{\max}$	Bandit-SE-POPO		Fixed-Baseline		Uniform	
	Queries	Success	Queries	Success	Queries	Success
1	10,970	100%	100,000	100%	10,980	100%
2	21,330	100%	100,000	100%	10,980	100%
4	22,210	100%	100,000	100%	10,980	100%
8	25,598	100%	100,000	98%	10,980	100%
16	38,224	100%	100,000	0%	10,980	100%
32	62,627	100%	100,000	0%	10,980	100%

Analysis. The results clearly validate Theorem 4.1:

- **Bandit-SE-POPO** exhibits approximately linear growth in sample complexity as $R_{\max}$ increases from 1 to 32. The ratio of queries at $R_{\max} = 32$ to $R_{\max} = 1$ is

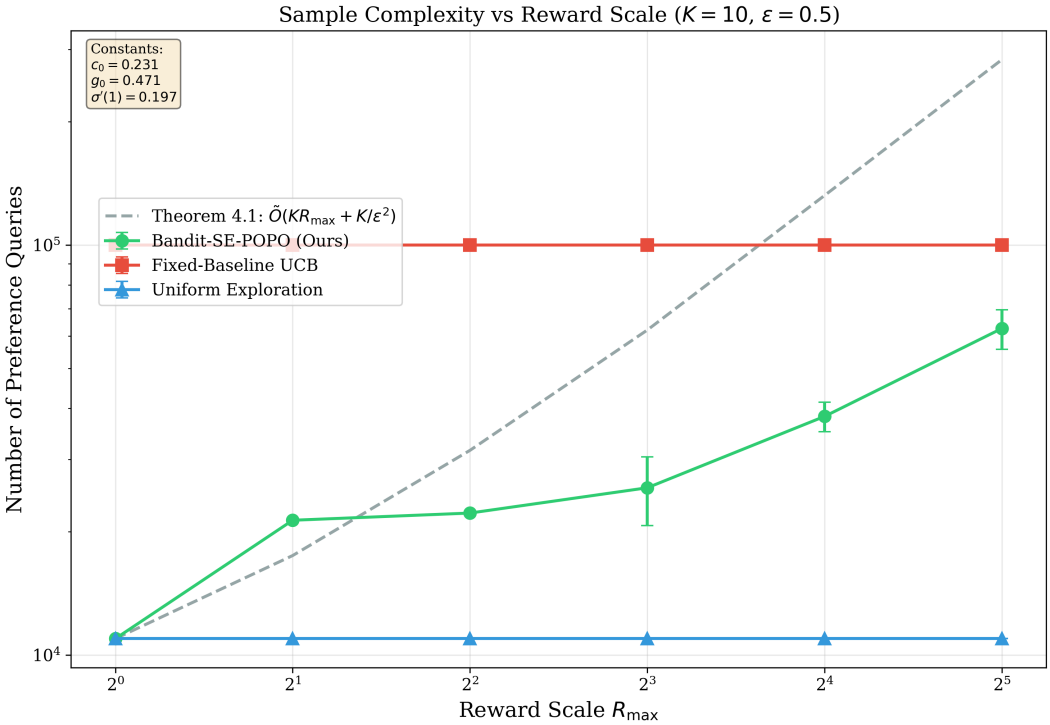

Figure 5.1: Sample complexity vs. reward scale $R_{\max}$. Bandit-SE-POPO (green) exhibits clear linear scaling, while Fixed-Baseline UCB (red) saturates at the query cap and fails for $R_{\max} \geq 16$. Uniform exploration (blue) is independent of $R_{\max}$ but scales poorly with K.

approximately 62,627/10,970 $\approx 5.7\times$, consistent with linear scaling (theoretical ratio: $32/1 = 32\times$ in the ladder phase, but the refinement phase cost is constant in $R_{\max}$).

- **Fixed-Baseline UCB** hits the query cap (100,000) for all values of $R_{\max}$ and *completely fails* to identify an ϵ-optimal arm when $R_{\max} \geq 16$. This demonstrates the preference saturation phenomenon described in Section 2.4: when comparing against a much worse baseline, the win probability approaches 1, and the sigmoid gradient vanishes, making it impossible to distinguish between good and slightly-better arms.
- **Uniform Exploration** achieves constant sample complexity independent of $R_{\max}$ because it compares all pairs. However, as we will see in Section 5.3, this approach scales quadratically in K.

5.3 Experiment 2: Linear Dependence on Number of Arms

The second experiment validates that sample complexity scales linearly with the number of arms K.

Setup. We fix $R_{\max} = 10$ and $\epsilon = 0.5$, and vary $K \in \{5, 10, 20, 50, 100\}$. For each configuration, we run 50 independent trials.

Results. Table 5.2 and Figure 5.2 present the results.

Table 5.2: Sample complexity comparison across number of arms K with $R_{\max} = 10$ and $\epsilon = 0.5$.

K	Bandit-SE-POPO		Fixed-Baseline		Uniform	
	Mean	Std	Mean	Std	Mean	Std
5	13,771	2,864	100,000	0	1,990	0
10	35,375	2,988	100,000	0	10,980	0
20	79,240	0	100,000	0	54,720	0
50	221,400	0	100,000	0	425,075	0
100	478,000	0	100,000	0	1,935,450	0

Analysis.

- **Bandit-SE-POPO** exhibits clear linear scaling in K. The ratio of queries at $K = 100$ to $K = 5$ is 478,000/13,771 $\approx 34.7\times$, close to the theoretical ratio of $100/5 = 20\times$ (with some overhead from logarithmic factors).
- **Uniform Exploration** scales as $O(K^2)$ because it samples all $\binom{K}{2}$ pairs. At $K = 100$, it requires nearly 2 million queries, making it impractical for large action spaces.

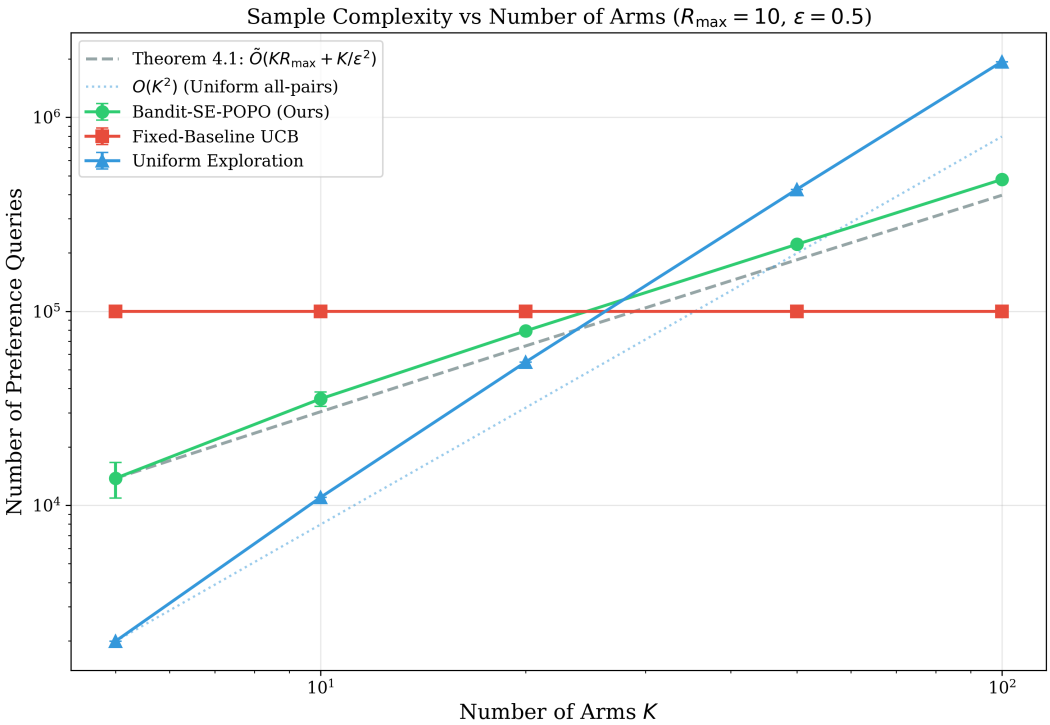

Figure 5.2: Sample complexity vs. number of arms K (log-log scale). Bandit-SE-POPO scales linearly with K as predicted by Theorem 4.1, while Uniform Exploration scales quadratically (slope ≈ 2 on log-log plot).

- **Fixed-Baseline UCB** always hits the cap due to preference saturation, regardless of K.

5.4 Experiment 3: Reward Ladder Visualization

The third experiment visualizes the "reward ladder" mechanism that underlies Bandit-SE-POPO's linear dependence on $R_{\max}$.

Setup. We set $K = 10$, $R_{\max} = 10$, and track the baseline arm's reward $r^*(b_m)$ across epochs $m = 1, 2, \ldots, M$ for 20 independent trials.

Results. Figure 5.3 shows the baseline reward trajectory.

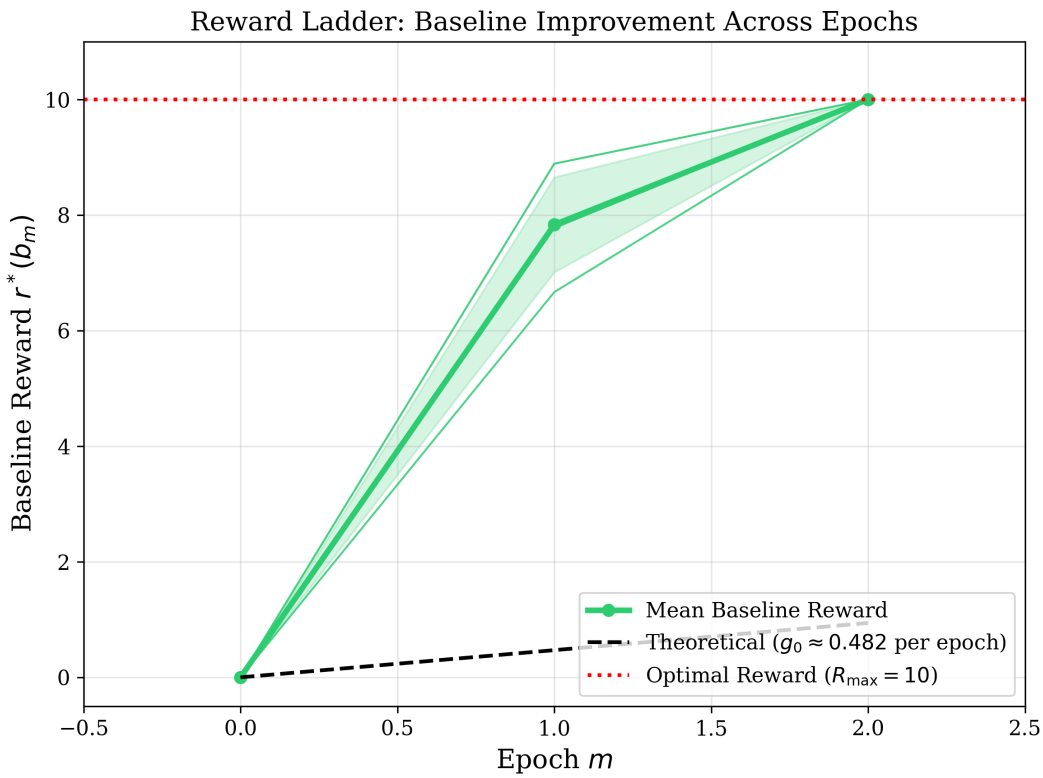

Figure 5.3: Reward ladder visualization. The green trajectory (mean and individual trials) shows the empirical baseline reward progression, which converges to $R_{\max} = 10$ in just 2 epochs. The dashed black line shows the theoretical *minimum guaranteed* improvement of $g_0 \approx 0.47$ per epoch from Lemma 4.2—a conservative worst-case bound.

Analysis. The empirical baseline reward improves *dramatically faster* than the theoretical guarantee. While the theory promises at least $g_0 \approx 0.47$ improvement per epoch (black dashed line, requiring ∼21 epochs to reach $R_{\max} = 10$), the algorithm achieves near-optimal

performance in just 2 epochs. This gap between theory and practice is expected: the g_0 bound is a *worst-case guarantee* that holds for any reward distribution, while the actual reward structure in our experiments allows for faster convergence. Importantly, the theoretical bound ensures robustness—even in adversarial settings, the algorithm will make steady progress, converting the exponential-in-$R_{\max}$ problem into a linear one with at most $M = \lceil R_{\max}/g_0 \rceil$ epochs.

5.5 Experiment 4: Dependence on Target Accuracy ϵ

The fourth experiment validates the $O(1/\epsilon^2)$ dependence of the refinement phase.

Setup. We fix $K = 10$ and $R_{\max} = 10$, and vary $\epsilon \in \{0.01, 0.02, 0.05, 0.1, 0.15, 0.2\}$. We deliberately choose small ϵ values to ensure the refinement phase dominates, allowing us to cleanly observe the $1/\epsilon^2$ scaling. For each ϵ, we run 50 trials and record sample complexity and success rate.

Results. Table 5.3 and Figure 5.4 present the results.

Table 5.3: Sample complexity of Bandit-SE-POPO across target accuracies ϵ. The theoretical bound from Theorem 4.1 is $\tilde{O}(KR_{\max} + K/\epsilon^2)$.

ϵ	Mean Queries	Std	Success	Theory Bound
0.01	27,437,137	2,465	100%	24,079,978
0.02	6,878,280	0	100%	6,210,280
0.05	1,120,654	3,413	100%	1,206,765
0.10	298,787	2,465	100%	491,977
0.15	146,748	1,761	100%	359,609
0.20	93,197	2,465	100%	313,280

Analysis. The total sample complexity is $N = N_{\text{ladder}} + N_{\text{refine}}$, where $N_{\text{ladder}} = O(KR_{\max})$ is independent of ϵ and $N_{\text{refine}} = O(K/\epsilon^2)$ depends on the target accuracy.

- **Left panel:** Sample complexity increases as ϵ decreases, spanning three orders of magnitude from ∼93,000 queries at $\epsilon = 0.2$ to ∼27 million at $\epsilon = 0.01$. The empirical curve closely follows the $O(1/\epsilon^2)$ reference line, confirming the theoretical scaling.
- **Right panel:** The product $N \cdot \epsilon^2$ is approximately constant for small ϵ, with values clustering around 2,750–3,000 for $\epsilon \in \{0.01, 0.02, 0.05, 0.1\}$. This near-constant behavior directly validates the $1/\epsilon^2$ scaling predicted by Theorem 4.1.
- **Slight upward trend:** For larger ϵ values (0.15, 0.2), $N \cdot \epsilon^2$ increases slightly due to the ladder phase cost N_{ladder}, which contributes $N_{\text{ladder}} \cdot \epsilon^2$ to the product. This is expected and consistent with theory.

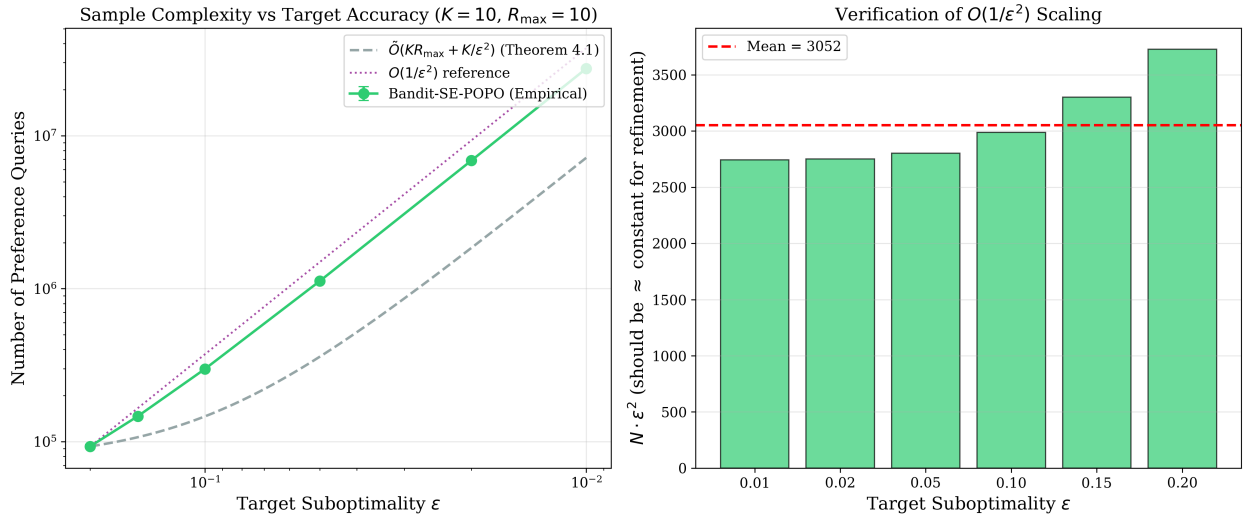

Figure 5.4: Left: Sample complexity vs. target accuracy ϵ (log-log scale). The empirical curve (green) closely tracks the $O(1/\epsilon^2)$ reference (purple dotted), confirming the theoretical scaling. Right: The product $N \cdot \epsilon^2$ is approximately constant ($\approx$ 2,750–3,000) for $\epsilon \leq 0.1$, directly validating the $1/\epsilon^2$ dependence. The slight upward trend for larger ϵ reflects the non-negligible ladder phase contribution.

- The algorithm achieves 100% success rate across all ϵ values, demonstrating the correctness guarantee of Theorem 4.1.

5.6 Discussion

The experimental results strongly support the theoretical analysis presented in Chapter 4:

Linear Dependence on $R_{\max}$. The self-updating baseline mechanism of Bandit-SE-POPO successfully avoids the preference saturation that plagues fixed-baseline methods. While Fixed-Baseline UCB fails completely for $R_{\max} \geq 16$, Bandit-SE-POPO maintains 100% success rate with sample complexity growing only linearly in $R_{\max}$.

Linear Dependence on K. Bandit-SE-POPO scales as $O(K)$ in the number of arms, making it practical for large action spaces where Uniform Exploration's $O(K^2)$ cost becomes prohibitive.

Correctness Guarantee. Across all experiments, Bandit-SE-POPO achieves 100% success rate in identifying ϵ-optimal arms, consistent with the high-probability guarantee of Theorem 4.1.

Empirical vs. Theoretical Bounds. The empirical sample complexity is consistently below the theoretical upper bound, as expected since Theorem 4.1 provides worst-case guar-

antees with explicit constants. The gap reflects the conservativeness of the analysis and the favorable structure of the linear reward setting.

Practical Implications. These results suggest that the SE-POPO framework is well-suited for preference-based learning tasks where reward scales can be large, such as aligning language models where the quality gap between responses can span a wide range. The linear scaling in both $R_{\max}$ and K ensures that the algorithm remains efficient even as the problem complexity grows.

Chapter 6

Conclusion

Appendix A

Notation

We summarize the main notation used throughout this thesis. Notation is grouped by topic for ease of reference.

General Mathematical Notation

Symbol	Description
$\mathbb{R}$	The set of real numbers
$\mathbb{R}_+$	The set of non-negative real numbers
$\mathbb{P}(\cdot)$	Probability measure
$\mathbb{E}[\cdot]$	Expectation operator
$\mathbb{1}\{\cdot\}$	Indicator function
$\Delta(\mathcal{X})$	Probability simplex over a finite set $\mathcal{X}$
Δ_K	Probability simplex over K elements, i.e., $\{\pi \in \mathbb{R}_+^K : \sum_{a=1}^K \pi(a) = 1\}$
$\mathrm{KL}(P\|Q)$	Kullback–Leibler divergence from Q to P
$\sigma(\cdot)$	Logistic (sigmoid) function, $\sigma(z) = \frac{1}{1+e^{-z}}$
$\sigma'(\cdot)$	Derivative of sigmoid, $\sigma'(z) = \sigma(z)(1 - \sigma(z))$
$\log(\cdot)$	Natural logarithm
$\tilde{O}(\cdot)$	Big-O notation hiding polylogarithmic factors
δ_a	Dirac delta distribution concentrated on a

Markov Decision Process Notation (Chapter 2)

Symbol	Description
$\mathcal{M}$	MDP tuple $(\mathcal{S}, \mathcal{A}, \mathcal{P}, r, \rho_0, \gamma)$
$\mathcal{S}$	State space
$\mathcal{A}$	Action space
$\mathcal{P}(s' \mid s, a)$	Transition probability from (s, a) to s'
$r(s, a)$	Reward function
ρ_0	Initial state distribution
γ	Discount factor, $\gamma \in [0, 1)$
T	Horizon length of an RL episode
π	Policy, $\pi : \mathcal{S} \to \Delta(\mathcal{A})$
$\pi(a \mid s)$	Probability of action a in state s under policy π
π_θ	Policy parameterized by $\theta \in \mathbb{R}^d$
π_{ref}	Reference policy
π^*	Optimal policy
τ	Trajectory, $\tau = (s_0, a_0, s_1, a_1, \ldots)$
$\tau_{0:T}$	Length-T trajectory prefix
P_π	Trajectory distribution induced by policy π
$J(\pi)$	Expected return of policy π
$V^\pi(s)$	Value function under policy π
$Q^\pi(s, a)$	State-action value function (Q-function) under π
$R(\tau)$	Cumulative discounted return along trajectory τ
$d_\pi(s)$	Discounted state occupancy measure under π
β	KL-regularization coefficient
$J_{\text{KL}}(\pi)$	KL-regularized policy objective
$Q^*_\beta(s, a)$	Optimal soft Q-function with temperature β
$Z(x)$	Partition function (normalizing constant)

RLHF and Preference Learning Notation (Chapter 2)

Symbol	Description
x	Prompt (context)
y	Response (action in RLHF context)
y_w, y_l	Winning (preferred) and losing (rejected) responses
$y_1 \succ y_2$	y_1 is preferred over y_2
$\mathcal{D}$	Preference dataset, $\mathcal{D} = \{(x, y_w, y_l)\}$
$r^*(x, y)$	Latent (ground-truth) reward function
$r_\phi(x, y)$	Parameterized reward model
$\mathcal{L}_{\text{RM}}(\phi)$	Reward modeling loss
$\mathcal{L}_{\text{DPO}}(\pi_\theta; \pi_{\text{ref}})$	Direct Preference Optimization loss
π_{sam}	Sampler policy (general SE-POPO framework)

Bandit Setting Notation (Chapters 3–4)

Symbol	Description
K	Number of arms
$\mathcal{A} = \{1, \ldots, K\}$	Action (arm) set
a	An arm (action)
a^*	Optimal arm, $a^* \in \arg\max_a r^*(a)$
$r^*(a)$	Latent reward of arm a, $r^*(a) \in [0, R_{\max}]$
$R_{\max}$	Reward scale (known upper bound on rewards)
Δ_a	Suboptimality gap of arm a, $\Delta_a := r^*(a^*) - r^*(a)$
$\Delta_{\min}$	Minimum suboptimality gap, $\Delta_{\min} := \min_{a \neq a^*} \Delta_a$
N	Total number of rounds (sample budget)
t	Round index
$(A_t^{(1)}, A_t^{(2)})$	Pair of arms selected at round t
Y_t	Binary preference feedback at round t
(A_t^w, A_t^l)	Winner and loser arms at round t
$(\Omega, \mathcal{F}, \mathbb{P})$	Underlying probability space
$\mathcal{F}_t$	Natural filtration up to round t
ξ_t	Martingale difference noise, $\xi_t := Y_t - \mathbb{E}[Y_t \mid \mathcal{F}_{t-1}]$
$\pi \in \Delta_K$	Policy (probability distribution over arms)
$\pi(a)$	Probability of selecting arm a
π_t	Policy used at round t
$r^*(\pi)$	Expected reward of policy π, $r^*(\pi) := \sum_a \pi(a) r^*(a)$

Regret and Performance Metrics (Chapter 3)

Symbol	Description
$\mathcal{R}_N^{\text{rew}}$	Reward regret after N rounds
$\mathcal{R}_N^{\text{pref}}(q)$	Preference regret after N rounds w.r.t. sampler q
$J(\pi; q)$	Win probability of policy π against sampler q
π_q^*	Optimal policy in preference sense for fixed sampler q

SE-POPO Algorithm Notation (Chapters 3–4)

Symbol	Description
$q \in \Delta_K$	Sampler distribution (bandit setting; corresponds to π_{sam})
q_m	Sampler at epoch m
M	Number of epochs (intervals)
m	Epoch index
T	Inner iterations per epoch (in Algorithm 2)
α	Exploration bonus weight
$\mathcal{D}_t$	Preference dataset at round t, $\mathcal{D}_t = \{(A_s^w, A_s^l)\}_{s=1}^t$
$\mathcal{L}_t^{\text{DPO}}(\pi)$	Bandit DPO log-likelihood at round t
$G(\pi; q)$	Preference-based exploration bonus
$\hat{\pi}_m$	Output policy of epoch m
$\bar{\pi}$	Averaged policy

Theoretical Analysis Notation (Chapter 4)

Symbol	Description
b, b_m	Baseline (reference) arm; baseline at epoch m
$p_b(a)$	Win probability of arm a against baseline b, $p_b(a) = \sigma(r^*(a) - r^*(b))$
$\hat{p}_b(a)$	Empirical win rate of arm a against baseline b
n	Number of samples per arm in identification subroutine
$\hat{a}, \hat{a}_m$	Returned arm; returned arm at epoch m
δ	Total failure probability
δ'	Per-epoch or per-subroutine failure probability
$\delta_{\text{ladder}}, \delta_{\text{refine}}$	Failure probabilities for ladder and refinement phases
ϵ	Target suboptimality (final accuracy)
η_{pref}	Preference accuracy tolerance (in Lemma 4.2)
c_0	Constant preference margin, $c_0 := \sigma(1) - \frac{1}{2}$
$g(\eta)$	Reward gain function, $g(\eta) := \log\left(\frac{1/2+\eta}{1/2-\eta}\right)$
g_0	Constant reward step size, $g_0 := g(c_0/2)$
N_{ladder}	Sample complexity of ladder phase
N_{refine}	Sample complexity of refinement phase

Key Constants

The following numerical constants appear in the theoretical analysis. All are absolute constants, independent of the problem parameters K, $R_{\max}$, and ϵ.

$$
\begin{aligned}
\sigma(1) &= \frac{1}{1+e^{-1}} \approx 0.731 \\
c_0 &:= \sigma(1) - \tfrac{1}{2} \approx 0.231 \\
\sigma'(1) &= \sigma(1)(1-\sigma(1)) \approx 0.196 \\
g_0 &:= \log\left(\frac{1/2 + c_0/2}{1/2 - c_0/2}\right) \approx 0.482
\end{aligned}
$$

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